

CURRICULUM VITAE

WON -YOUNG CHOI

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RESEARCH AREAS

Spintronics, Spin-orbit torques, Spin transport, Magnetic materials

EDUCATION

Mar. 2023 – **Ph.D. in Material Science and Engineering | Seoul National University**, Seoul, Korea
Present • Advisor: Prof. Hyejin Jang, Co-advisor: Dr. Dong-Soo Han (KIST)

Mar. 2019 – **M.S. in Physics | Sogang University**, Seoul, Korea
Feb. 2021 • GPA 3.76 / 4.3
• Academic excellence scholarship for freshmen (1st semester, 2019)
• Leader of the thin film subgroup (1st semester, 2020)
• Master thesis was replaced by the published SCI(E) Journal paper (publication No. 11)
• Advisor: Prof. Myung-Hwa Jung

Mar. 2012 – **B.S. in Physics | Sogang University**, Seoul, Korea
Feb. 2019 • Graduated Summa Cum Laude (3 of 24)
• GPA 3.78/4.3 & Major GPA 4.09/4.3
• Honors scholarship (2nd semester, 2016 & 1st and 2nd semester, 2017)
• Dean's list (overall 2017 semesters)

RESEARCH & WORK EXPERIENCES

Mar. 2023 – **Student Researcher | Center for Semiconductor Technology, Korea Institute of Science and Technology (KIST)**, Seoul, Korea
Present

- Oct. 2021 – **Research Intern | Center for Spintronics, Korea Institute of Science and Technology (KIST), Seoul, Korea**
Feb. 2023
- Mar. 2021 – **Visiting Researcher | Kläui Lab, Johannes Gutenberg-Universität Mainz (JGU), Mainz, Germany**
Aug. 2021
- Mar. 2019 – **Teaching Assistant | Sogang University, Seoul, Korea**
Jun. 2020
- Experimental Physics 1 class (1st semester, 2019 and 2020)
 - Experimental Physics 2 class (2nd semester, 2019)
 - Mathematical Physics 1 class (1st semester, 2019)
- Mar. 2018 – **Undergraduate Student Researcher | Sogang University, Seoul, Korea**
Feb. 2019
- Fabricated magnetic thin film using magnetron co-sputtering
 - Evaluated magnetic properties of magnetic thin film using MPMS-3 (SQUID-VSM)
 - Investigated electrical properties of magnetic thin film
- Oct. 2013 – **Air Materials Supply Sergeant | 15th Special Missions Wing, Republic of Korea Air Force, Seoul, Korea**
Oct. 2015
- Served military service
 - Chief soldier (Jan. 2015 – Jul. 2015)

PUBLICATIONS & PRESENTATIONS

PUBLICATIONS († co-first author, *corresponding author)

1. J.-B. Choi, H. Seong, **W.-Y. Choi**, S.-J. Noh, J.-S. Lee, H.-S. Lee, Y.-C. Kim, O.-J. Lee, B.-C. Min, J.-H. Kwon*, D.-S. Han*, Spin Hall Magnetic Sensors with Dynamic Offset Calibration and Structurally Tunable Sensitivity. [IEEE Sensors Journal 25, 21, 39605-39611 \(2025\)](#).
2. **W.-Y. Choi**†, J.-H. Ha†, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong*, K.-W. Kim*, D.-S. Han*, Magnetization switching driven by magnonic spin dissipation. [Nature Communications 16, 5859 \(2025\)](#).
3. S. Yu, J. Park, Y. Jo, W. Kim, H. Kim, **W.-Y. Choi**, M. Kim, D.-S. Han, M.-H. Jung, K. Rhie, K. Lee*, The control of magnetization reversal by tuning the interlayer exchange interaction in magnetic multilayers. [Journal of the Korean Physical Society 86, 62-67 \(2024\)](#).
4. H. Seong, **W.-Y. Choi**, J. Choi, D.-H. Kim, T.-E. Park, B.-C. Min, H.-J. Choi, D.-S. Han*, Magnetic properties of GdFeCo thin films tailored by sputtering conditions. [Current Applied Physics 68, 131-137 \(2024\)](#).

5. S. Jena, R. Urkude, **W.-Y. Choi**, K. K. Pandey, S. Karwai, M. H. Jung, J. Gardner, B. Ghosh, V. R. Singh*, Electronic Structures of Skyrmionic Polycrystalline MnSi Thin Film Studied by Resonance Photoemission and X-ray Near Edge Spectroscopy. [*Journal of Applied Physics* **135**, 165301 \(2024\).](#)
6. S. Jena*, R. Dawn, **W.-Y. Choi**, Y. Singh, A. Chosh, S. K. Sahoo, M. H. Jung, J. Gardner, V. K. Verma, K. Amemiya, V. R. Singh*, Texture, Electronics and Magnetic States of Mn Atoms in Polycrystalline MnSi Thin Films Facing ac-Sapphire Substrate Studied by Soft X-ray Magnetic Circular Dichroism. [*Thin Solid Films* **786**, 140097 \(2023\).](#)
7. F. Kammerbauer, **W.-Y. Choi**, F. Freimuth, K. Lee, F. Robert, D. -S. Han, H. J. M. Swagten, Y. Mokrousov, M. Kläui*, Controlling the Interlayer Dzyaloshinskii–Moriya Interaction by Electrical Currents. [*Nano letters* **23**, 15, 7070-7075 \(2023\), *arXiv*: 2209. 01450.](#)
8. S. Jena, **W.-Y. Choi**, J. Gardner, M. H. Jung, S K Srivastava, V K Verma, K Amemiya, V R Singh*, Evolution of bulk magnetic structure in MnSi thin film: a soft x-ray magnetic circular dichroism study. [*Physica Scripta* **98**, 075927 \(2023\).](#)
9. **W.-Y. Choi**, W. Yoo, M.-H. Jung*, Emergence of the topological Hall effect in a tetragonal compensated ferrimagnet Mn_{2.3}Pd_{0.7}Ga. [*NPG Asia Materials* **13**, 79 \(2021\).](#)
10. **W.-Y. Choi**[†], J. H. Jeon[†], H. W. Bang, W. Yoo, SK. Jerng, S. H. Chun*, S. Lee*, M.-H. Jung*, Proximity-Induced Magnetism Enhancement Emerged in Chiral Magnet MnSi/Topological Insulator Bi₂Se₃ Bilayer. [*Advanced Quantum Technologies* **4**, 2000124 \(2021\).](#)
11. **W.-Y. Choi**, H.W. Bang, S. H. Chun, S. Lee, M.-H. Jung*, Skyrmion Phase in MnSi Thin Films Grown on Sapphire by a Conventional Sputtering. [*Nanoscale Research Letters* **16**, 7 \(2021\).](#)

PRESENTATIONS

1. **W.-Y. Choi**, J.-H. Ha, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Self-generated Spin-orbit Torques Driven by Magnonic Spin dissipation in Antiferromagnet/Ferromagnet Bilayer. 2025 70th MMM Annual Conference [*Invited Talk*](#).
2. **W.-Y. Choi**, J.-H. Ha, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Magnonic Spin Dissipation Driven Anisotropic Spin Torque in Easy-Axis Antiferromagnet/Ferromagnet Bilayer. 2025 KMS Summer Conference [*poster presentation*](#).
3. **W.-Y. Choi**, H.-J. Ha, M.-S. Jung, S. Kim, O. Lee, B.-C. Min, M. Kläui, H. Jang, J.-I. Hong, K.-W. Kim, D.-S. Han, Spin-orbit torque by magnon dissipation. 2023 MML Conference [*poster presentation*](#).

4. **W.-Y. Choi**, H.-J. Ha, H. Jang, B.-C. Min, J.-I. Hong, K.-W. Kim, D.-S. Han, Spin Hall magnetoresistance and unidirectional magnetoresistance in ferromagnet metal/antiferromagnet insulator bilayers. 2023 KMS Summer Conference poster presentation.
5. **W.-Y. Choi**, H.-J. Ha, M.-S. Jung, B.-C. Min, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Spin-orbit torque by magnon dissipation. 2022 KMS Winter Conference poster presentation.
6. **W.-Y. Choi**, H. W. Bang, Y. C. Park, M.-H. Jung, Topological Hall effect in a mixed phase of Mn_3Ga thin film. 2022 KMS Summer Conference poster presentation.
7. **W.-Y. Choi**, W. Yoo, M. H. Jung, Observation of unconventional anomalous Hall effect at compensated $\text{Mn}_{2.3}\text{Pd}_{0.7}\text{Ga}$ thin film. 2020 KPS Fall Meeting oral presentation.
8. **W.-Y. Choi**, W. Yoo, H.W. Bang, M.T. Park, K. Lee, M.-H. Jung, Compensated ferrimagnetism of $\text{Mn}_{2+\delta-x}\text{Pd}_x\text{Ga}$ thin film. 2019 KPS Fall Meeting poster presentation.

HONORS & AWARDS

Jul. 2026	STAR Outstanding Graduate Student Award , The Korean Magnetism Society (KMS) 2026
Dec. 2025	PSI Excellent Student Researcher Award , Korea Institute of Science and Technology Post-Silicon Semiconductor Institute 2025
Oct. 2025	Condensed Matter Physics MSM Awards , The Korean Physical Society (KPS) 2025 Fall Meeting
Nov. 2022	Best Poster Award , The Korean Magnetism Society (KMS) 2022 Winter Conference
May. 2022	Poster Award , The Korean Magnetism Society (KMS) 2022 Summer Conference
Jan. 2021	International Academic Papers Scholarship (\$500) , Ministry of Education, Korean government
Oct. 2020	Outstanding Presentation Award , The Korean Physical Society (KPS) 2020 Fall Meeting

- Mar. 2019 **Academic Excellence Scholarship for Freshmen (\$3200)**, Graduate School of Sogang University, Seoul, Korea
- Mar. 2018 **CHO-LEE Scholarship (\$1000)**, 1st semester, 2018, Corporation CHO-LEE, Seoul, Korea
- Jan. 2018 **Dean's List Award**, Sogang University, Seoul, Korea (Award for Top 3% Student)
- Sept. 2016 – Dec. 2017 **Honors Scholarship (\$2100)**, 2nd semester, 2016, and 1st & 2nd semester, 2017, Sogang University, Seoul, Korea

PERSONAL SKILLS

- **Sample Preparation Techniques:** Magnetron Sputtering, Photolithography, Ar ion milling, Wire Bonder
 - **Measurements Techniques:** Scanning Electron Microscopy (SEM), Energy dispersive X-ray spectroscopy (EDXS), X-ray diffraction (XRD), Van der Pauw Method, Magnetic Properties Measurement System (MPMS), Physical Properties Measurement System (PPMS), Electrical Transport Properties Measurement, 2nd Harmonics Spin-orbit torque Measurement
 - **Programming Language:** C, Python, Labview
 - **Software:** Origin Lab, Qtiplot, VESTA, ElmerFEM, NanoCAD, FreeCAD, Illustrator, Blender
 - **Languages:** Native Korean, Fluent English, Intermediate Japanese
 - **Sports:** Kendo (2nd dan)
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